

Assignment 7

Secure Network Application Development Using TCP

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1. Objective

Enhance the GUI-based multi-client TCP application developed in Assignment 6 by implementing practical security mechanisms. The emphasis is on application security—including authentication, password protection, session management, and input validation—while reusing the existing networking architecture and graphical interface.

2. Security Features Implemented

- User Authentication: Requires clients to provide a valid username and password to join the chat.
- Secure Password Storage: Passwords are never stored or displayed in plaintext; they are hashed using the SHA-256 algorithm before being saved.
- Duplicate Login Prevention: Tracks active sessions in memory to prevent the same username from being logged in concurrently.
- Input Validation: Rejects invalid username formats, empty passwords, unsupported commands, and oversized messages (exceeding 2048 bytes).
- Failed Login Protection: Implements a temporary 60-second account lockout after five consecutive failed login attempts.
- Session Management & Secure Logging: Disconnects users after 5 minutes of inactivity and securely logs security events without exposing credentials.

3. System Architecture

The application maintains the separation of concerns from Assignment 6. The networking layer (ChatClient) reuses the existing socket logic with an AUTH_REQUIRED and AUTH_SUCCESS/AUTH_FAIL handshake. The tkinter LoginWindow and ChatWindow were extended with a masked password field. The server validates credentials from users.csv before allowing chat access.

4. Implementation

Cryptography uses hashlib.sha256() for password hashing. User credentials are stored in users.csv with hashes only. Security events are appended to security_log.txt with timestamps, usernames and IP addresses while excluding password information.

5. Testing

Test Scenario	Expected Outcome	Result
Account Registration & Password Hashing	SHA-256 hash stored in users.csv	Passed
Successful Authenticated Login	User enters chat; LOGIN_SUCCESS logged	Passed
Duplicate Login Attempt	Second login rejected	Passed
Failed Login / Account Lockout	60-second lockout after 5 failures	Passed
Inactivity Timeout	Disconnect after 5 minutes	Passed

6. Wireshark Verification

TCP Handshake

Captured SYN, SYN-ACK and ACK packets establishing the connection before authentication.

Successful Login

TCP stream shows AUTH_REQUIRED followed by AUTH_SUCCESS and welcome banner.

Failed / Duplicate Login

Server responds with AUTH_FAIL: User already logged in and terminates the connection.

Session Timeout

Server sends an inactivity message followed by FIN, ACK to close the socket.

7. Conclusion

The secured multi-client chat application successfully integrates SHA-256 password hashing, authentication, active session tracking, account lockout, secure logging, and input validation while preserving the Assignment 6 architecture. The implementation was verified using Mininet, server logs, client testing, and Wireshark packet analysis.